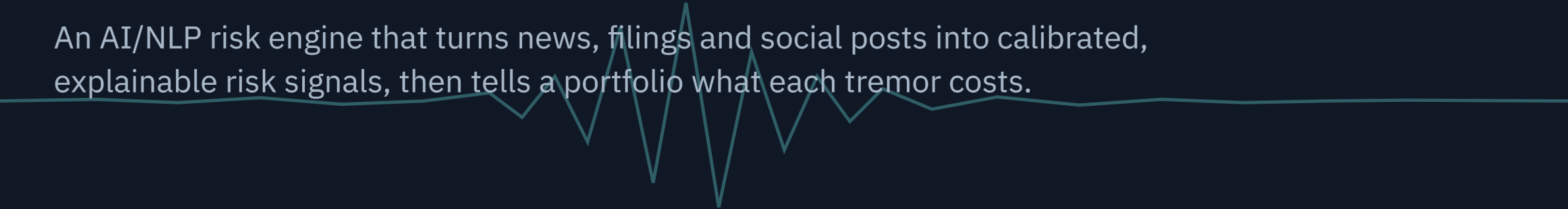


S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Seismo measures the magnitude of market-moving news

An AI/NLP risk engine that turns news, filings and social posts into calibrated, explainable risk signals, then tells a portfolio what each tremor costs.



Risk now moves at social-media speed; risk systems do not

\$42bn

withdrawn from Silicon Valley Bank on 9 March 2023, in a run organised in founders' group chats

\$589bn

lost by Nvidia in one session on 27 January 2025, after a weekend of coverage of a cheaper AI model

1 fake

image of a Pentagon explosion, from an impersonating account, briefly moved the S&P 500 in May 2023

THE APPROACH

Score every story per company, calibrate its impact on real market reactions, let only corroborated events trigger action, and feed both a tactical index overlay and a bank stress test.

One signal bus feeds a tactical and a strategic consumer

Seismo: one signal bus feeds a tactical and a strategic consumer

Replay packs and live sources share every downstream step. Default run is offline and deterministic; live mode is opt-in.

GDELT news

DOC 2.0 API, 15-min, keyless

SEC EDGAR 8-K

item codes as authoritative labels

Bluesky Jetstream

real-time posts, cashtags, author caps

Replay packs

SVB, DeepSeek, tariff, red team, quiet day (JSONL)

↓ normalize: one Document schema, UTC timestamps, content-hash ids, live or replay clock

AI/NLP Risk Engine

CPU-only, deterministic in replay, ~1 ms per document

1 Event classifier

8-K items + weighted rules, 10 classes, subtype, confidence

2 Entity linking

aliases, cashtags, CIK; context check for "Apple", "Meta"; relevance 0-100

3 Entity sentiment

clause-level window per entity; FinBERT or finance lexicon; -1..+1, confidence

4 Credibility

by domain and owner, never by badge; lookalike domains flagged

5 Event clusters

MinHash syndication, 24h novelty chains, coordinated-repost groups, denials → retraction

6 Impact 1-10

8-K event study: $P(|\text{SCAR}| > 2)$, isotonic-calibrated; "how big if true"



Signal bus · schema v1.1

document, event and entity grains · evidence spans are exact substrings · SQLite store · FastAPI REST + WebSocket · JSON Schema

Module A · tactical rebalancer

20 S&P 100 names, 11 sectors · S&P DJI-style filters · decayed, shrunk z-scores · inverse-vol tilt on capped cap weights · 15% name / 5% active caps · circuit breaker · 2% no-trade band · 5 bp costs

Trigger gate

impact ≥ 8 · analog exists · confidence ≥ 0.7 · ≥ 3 independent publishers in 60 min or 1 authoritative · not disputed → TRIGGER / REVIEW / RETRACT

Module B · strategic stress test

trade blotter → wholesale book · historical-analog factor shocks · bonds, swaps, CDS, FX, options · Vasicek PD, migration, ECL staging · CET1 vs Basel 7% / RBI 8% · reverse stress · memo



Dashboard (Streamlit)

Risk Radar · Module A · Module B · Model Lab · Try It (type a headline, run the red team)

Runs offline from replay packs with no API keys; live GDELT, SEC EDGAR and Bluesky adapters share the same pipeline and the same code paths.

Four choices make the signals trustworthy

Sentiment per company, not per article

Each company is scored on its own clause, so "Microsoft gains as Google loses" gives one up and one down. Linking checks context: "Apple pie" never becomes AAPL.

Events, not posts

Syndicated copies keep the original voice, sister outlets count once, and five or more near-identical accounts collapse into one coordinated voice. Novelty follows 24-hour chains.

Impact calibrated on real reactions

Impact is the chance of a two-sigma abnormal move, learned from 3,145 SEC 8-K events with a market-model event study. No language model sets the number.

A five-check trigger gate

Impact 8+, a historical analog, confidence 0.7+, three independent publishers or one authoritative source, and no official denial. Anything else goes to human review.

Seismo beats naive baselines and called every replay correctly

Replay	Naive fires	Seismo triggers	When, or what happened
SVB 2023	4	1	25 h before the bank was closed
DeepSeek 2025	4	1	3 h before Monday's open
Tariff shock 2025	3	2	15 h before the first open
Red-team fake	1	0	held for review, then retracted
Quiet day	1	0	stayed calm

Naive pipeline: fire a stress test on any document with $10 \times |\text{sentiment}|$ above 7. Entity linking precision 1.00 vs 0.83 for exact name matching; about 1 ms per document on CPU.

0.77 vs 0.73

impact AUC for a two-sigma move, 2022–23 test set, vs the hand-set 8-K severity ranking

2.84σ vs 0.75σ

average abnormal move after the top vs the bottom impact decile

19% vs 180%

Module A turnover vs a naive tilt (DeepSeek); worst drawdown 2.76% vs 3.11% for the benchmark

Every number regenerates with `make results` and `make impact`. Replay packs are labelled synthetic reconstructions of public timelines; prices, rates and 8-K filings are real public data.

It slots into the news-to-capital chain banks already run

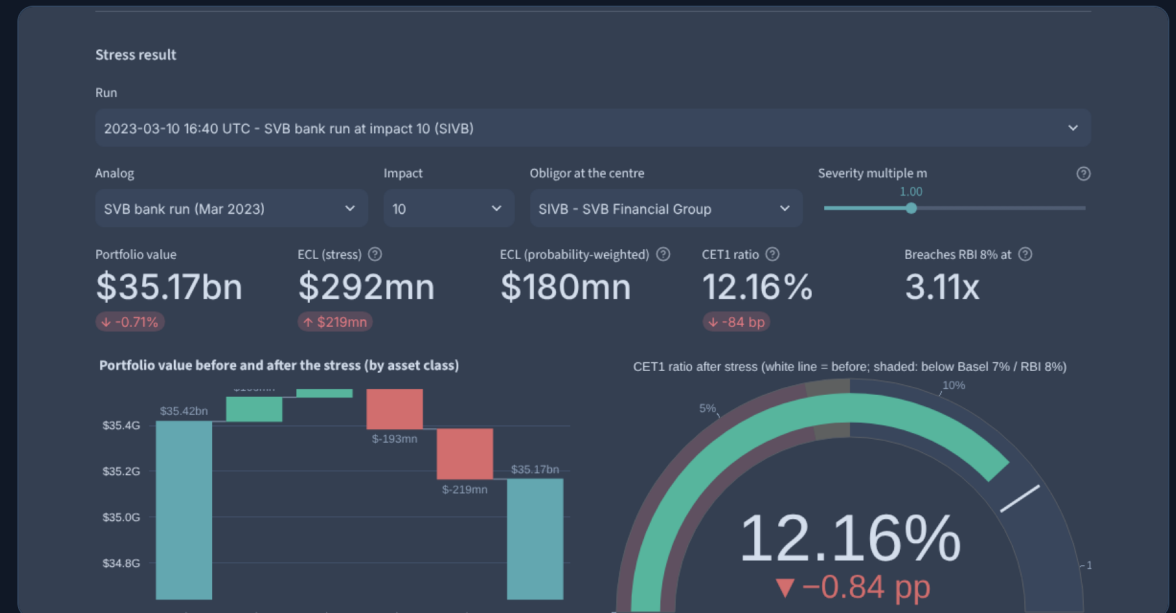


SVB ANALOG, IMPACT 10

CET1 13.00% → 12.16%

Expected credit loss rises from \$73mn to \$292mn as the bank at the centre defaults (Stage 3). The book breaches RBI's 8% floor at 3.1× the SVB shock; the tariff analog breaches at 1.9×.

Built for RBI's expected-credit-loss directions, effective 1 April 2027: three-stage ECL, probability-weighted scenarios, CET1 against Basel 7% and RBI 8%. Every number in the auto-written memo comes from the engine.



Synthetic wholesale book (225 positions from a seeded trade blotter). Shocks are each factor's largest move in the crisis window, computed from Yahoo Finance and FRED data.

What it does not do yet, and what comes next

LIMITS, STATED PLAINLY

- Archived 2023–25 social posts are not free, so the replays are labelled reconstructions; live Bluesky data is real
- Default sentiment is a finance lexicon; FinBERT is optional and not yet fine-tuned
- 8-K calibration covers company events; macro and geopolitical impact use a documented prior
- 20 names with today's index membership: survivorship bias

NEXT MONTH

- Target-masked FinBERT fine-tuned on SEntFiN and FiQA
- Hindi and regional-language news for an India view and a Nifty 50 universe
- Supply-chain spillovers and counterparty exposure
- Point-in-time constituents and a multi-year news backtest